

Genetic report for longevity

Muscle, Bone and Sarcopenia

Strength, lean mass, bone density, fractures, joints and tendons

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This report combines the genetics of muscle (strength, potency, lean mass, type of fiber), bone (mineral density, fractures, via Wnt/sclerostine and RANKL/OPG) and joints and tendons, focusing on preventing sarcopenia, osteoporosis and frailty.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

35

variants

26

in the chip

8

imputed
($DR2 \geq 0.8$)

1

not found

10

Scores with
percentile

Overall reading. The weakest point is muscle mass: the appendicular lean mass score is in percentile 9.5 and that of grip strength in the 29.5 (typical, but below average). This agrees with the fragility and sarcopenia "more likely" of a previous report. In bone, the genetic mineral density is typical (the 43rd percentile) and osteoporosis is also (42), but the fracture score is in the 84th percentile. The joints are a force: osteoarthritis in the 16th percentile and GDF5 without risk alleles. In classical genes, ACTN3 is RX (intermediate), PPARGC1A is Gly/Gly (good response to aerobic training), AMPD1 is normal and COL1A1 does not have the allele "s". The genes of muscle and bone respond a lot to the training of strength, protein and vitamin D: genetics says where to put the effort, not the result.

Favourable

- **Osteoarthritis:** the 16th percentile
- **PPARGC1A rs8192678 CC:** Gly482Ser: T allele (482Ser) lowers the activity of PGC-1 α (mitochondrial biogenesis), is associated with lower VO2max gain with training and higher risk of 2 type diabetes
- **AMPD1 rs17602729 GG:** Q12X (C34T): the allele A (strand +) creates a stop codon in muscle deaminase AMP
- **ZBTB40 rs6684375 TC:** 1p36 locus (ZBTB40/WNT4) of mineral density.
- **AXIN1 rs9921222 CC:** AXIN1: negative regulator of β -catenin (via Wnt).
- **WLS rs2566752 CC:** Wntless: secretion of Wnt proteins.
- **GDF5 rs224333 AA:** GDF5: cartilage and joint growth factor; the most replicated locus of osteoarthritis.

Unfavourable

- **Appendicular lean mass:** the 10th percentile
- **Fractures (in the last 5 years):** the 84th percentile
- **RSPO3 rs7741021 AA:** R-spondine 3: potentiates the Wnt pathway.
- **SMAD3 rs28587205 TT:** SMAD3: TGF- β pathway in bone and matrix.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 26 chip genotypes, 0 needed complement, 0 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Polygenic scores update (03-10-2026). The scores were recalculated with more published recent scores (the same method and the same reference population). This box prevails over the previous values cited in the text.

- Lean mass: new p32 (genomic) and p3,5 scores; with the previous ones (**p27**, p30 and p10), the consensus is p27. Under the average, lower than the p10 cited; strength and protein training remain.
- Fractures: the only new score (45 SNPs) gives p16, against p84 in the previous one; the genetic risk of fracture becomes uncertain and bone densitometry decides.

Strength, potency and muscle mass

Structural and muscle signaling genes (α -actinina-3, myostatin, PGC-1 α , AMPD1, ECA) and polygenic grip strength scores, appendicular lean mass, gait speed and sarcopenia. Handgrip strength and lean mass are the best predictors of advanced age mortality.

Ageing mechanisms: Sarcopenia; Thin Mass Loss; Loss of muscle strength; Muscular atrophy; Neuromuscular Junction Disorder; Fragility; Oxidative Phosphorylation Decline

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Handgrip strength (mean of the two hands)	Typical / intermediate	the 30th percentile · z=-0.51	Higher score = more strength. PGS001927 (PGS Catalog; Penalized regression (bigstatsr); Hand grip strength (mean of both); PGP000263, PMID 34995502). Variants with genotype: 71148 of 73846 (covering of 96.3% coefficients).
Appendicular lean mass	Unfavourable	the 10th percentile · z=-1.30	Higher score = more lean mass in the limbs (630 significant genomic variants). PGS005315 (PGS Catalog; Genome-wide signifier SNPs; Appendicular lean mass; PGP000766, PMID 38632276). Variants with genotype: 561 of 630 (covering of 89.6% coefficients).
Normal gait speed	Typical / intermediate	the 58th percentile · z=0.18	Higher score : faster gait (predictor of frailty and mortality). PGS001075 (PGS Catalog; snpnet; Usual walking pace; PGP000244, PMID 35324888). Variants with genotype: 6688 of 7531 (covering of 90.2% coefficients).
Sarcopenia (low grip strength at ≥ 60 years, EWGSOP/FNIH)	Typical / intermediate	the 50th percentile · z=-0.07	Higher score : higher risk of low strength in advanced age. Loci score of GWAS Catalog ($p \leq 510^{-8}$; 1 variant by 500 kb window; ln(OR) or beta coefficients): 18 of 19 loci with genotype. Studies: Jones G (Low grip strength (60 years and older) (EWGSOP)); Jones G (Low hand grip strength (60 years and older) (FNIH)). Variants with genotype: 18 of 19 (covering of 94.0% coefficients).
Frailty index	Typical / intermediate	the 69th percentile · z=0.53	Higher score = more predicted frailty. Loci score of GWAS Catalog ($p \leq 510^{-8}$; 1 variant by 500 kb window; ln(OR) or beta coefficients): 13 of 14 loci with genotype. Studies: Atkins JL (Frailty index). Variants with genotype: 13 of 14 (covering of 92.1% coefficients).

Favourable

- PPARGC1A rs8192678 CC
- AMPD1 rs17602729 GG

Unfavourable

- Appendicular lean mass (the 10th percentile)

Favourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
PPARGC1A	rs8192678 4:23814039 · C/T	CC	chip	T (0; 36%)	moderate	Gly482Ser: T allele (482Ser) lowers the activity of PGC-1 α (mitochondrial biogenesis), is associated with lower VO2max gain with training and higher risk of 2 type diabetes. Gene in strand – C in strand + = Gly.
AMPD1	rs17602729 1:114693436 · G/A/T	GG	chip	A (0; 12%)	moderate	Q12X (C34T): the allele A (strand +) creates a stop codon in muscle deaminase AMP. Carriers of 2 copies have deficiency of myoadenilate deaminase (crabs, early fatigue in intense effort); 1 copy has small effect.

Typical / intermediate variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
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frequency in reference)						
ACTN3	rs1815739 11:66560624 · C/A/T	CT	chip	T (1; 43%)	moderate	R577X: the T allele (577X) creates a stop codon; in XX it lacks α -actinina-3 in the fast fibers (II type). XX is associated with less power, more eccentric muscle damage and, in the elderly, more loss of strength and falls; RX is intermediate. Strand +: C = R, T = X.
FTO	rs1421085 16:53767042 · T/C	CT	chip	T (1; 57%)	weak	FTO: the adiposity locus also appears in lean mass and strength GWAS. Meaning: GWAS Catalog: Leg lean mass (Zhang YX 2020; p=0e+00); allele T ↓

Non-judgmental variants (informative) (5)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MSTN	rs1805086 2:190060351 · T/C	TT	chip	—	weak	K153R (myostatin): the C allele (153R) is rare; studies on mass and strength are small and contradictory. Informative, without judgment.
ACE	rs4343 17:63488670 · G/A	AG	chip	—	weak	Proxy of polymorphism I/D (G $\approx$ D allele, r2 $\approx$ 0.9). D: more ECA activity and force/power profile; I: aerobic resistance profile. No good/bad judgment; telmisartan acts downstream (AT1 receiver).
IGF1	rs35767 12:102481791 · A/C/G/T	GG	chip	—	weak	IGF1 promoter: T allele is associated with slightly higher circulating IGF-1. High IGF-1 helps the muscle but shortens life in animal models (GH/IGF-1 axis); therefore, without judgment.
CKM	rs8111989 19:45305950 · T/C	TT	chip	—	weak	Muscle creatine kinase (3'UTR): G allele was associated with resistance profile in some studies of athletes; inconsistent results. Information.
HIF1A	rs11549465 14:61740839 · C/T	CC	chip	—	weak	P582S: the T allele (582S) makes the HIF-1 α more stable and is associated with the power profile and faster fibers in small studies. Information.

Bone density, fractures and osteoporosis

Type I collagen, via Wnt (LRP5, sclerostine/SOST, WNT16, DKK1, RSPO3), RANKL/OPG axis (TNFSF11, TNFRSF11A, TNFRSF11B), estrogen receptor and vitamin D receptor. Polygenic scores of bone mineral density (heel), osteoporosis and fractures summarize hundreds of loci.

Ageing mechanisms: Osteoporosis; Fragility; Remodelling of the extracellular matrix; Decline of the Estrogen Signal; Endocrine dysregulation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Heel mineral density (ultrassom)	Typical / intermediate	the 43rd percentile · z=-0.23	Higher score = denser bone. PGS001955 (PGS Catalog; Penalized regression (bigstatsr); Heel bone mineral density (BMD); PGP000263, PMID 34995502). Variants with genotype: 70769 of 73565 (covering of 96.3% coefficients).
Osteoporosis	Typical / intermediate	the 42nd percentile · z=-0.20	Higher score : higher risk. PGS001883 (PGS Catalog; Penalized regression (bigstatsr); Osteoporosis; PGP000263, PMID 34995502). Variants with genotype: 361 of 365 (covering of 96.9% coefficients).
Fractures (in the last 5 years)	Unfavourable	the 84th percentile · z=0.99	Higher score : higher risk of fracture. PGS001921 (PGS Catalog; Penalized regression (bigstatsr); Fractured/broken bones in last 5 years; PGP000263, PMID 34995502). Variants with genotype: 7749 of 7965 (covering of 97.4% coefficients).

Favourable

- ZBTB40 rs6684375 TC
- AXIN1 rs9921222 CC
- WLS rs2566752 CC

Unfavourable

- Fractures (in the last 5 years) (the 84th percentile)
- ESR1 rs9340799 AA
- RSPO3 rs7741021 AA
- SMAD3 rs28587205 TT

Favourable variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ZBTB40	rs6684375 1:22379941 · C/T	TC	chip	C (1; 84%)	moderate	1p36 locus (ZBTB40/WNT4) of mineral density. Meaning: GWAS Catalog: Estimated mineral density (Chai RC 2026; p=6e-163); C ↓ allele
AXIN1	rs9921222 16:325782 · C/T	CC	chip	T (0; 42%)	moderate	AXIN1: negative regulator of β-catenin (via Wnt). Meaning: GWAS Catalog: Estimated mineral density (Chai RC 2026; p=6e-46); C ↑ allele
WLS	rs2566752 1:68191014 · T/A/C	CC	imputation DR2=0.98	T (0; 61%)	moderate	Wntless: secretion of Wnt proteins. Meaning: GWAS Catalog: Estimated mineral density (Chai RC 2026; p=8e-84); T ↓ allele

Unfavourable variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ESR1	rs9340799 6:151842246 · A/G	AA	chip	A (2; 69%)	weak	Xbal: the genotype XX (GG) was associated with fewer fractures in GENOMOS (Ioannidis 2004); the allele "x" (A) is the most common. Weak evidence.
RSPO3		AA	imputation DR2=1.00	A (2; 57%)	moderate	R-Spondin 3: potentiates the Wnt pathway. Meaning: GWAS Catalog: Estimated

rs7741021
6:127147129
· A/C

mineral density (Chai RC 2026; p=0e+00);
allele A ↓

SMAD3	rs28587205 15:67135559 · A/T	TT	imputation DR2=0.91	T (2; 46%)	moderate	SMAD3: TGF-β pathway in bone and matrix. Meaning: GWAS Catalog: Heel bone mineral density (Kemp JP 2017; p=3e-10); allele A ↑
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Typical / intermediate variants (17)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
COL1A1	rs1800012 17:50200388 · C/A	CC	chip	A (0; 19%)	moderate	Sp1 (G>T in the gene; A na strand +: the "s" allele changes the proportion of α1/α2 collagen type I chains and makes the bone more fragile; ss is associated with less mineral density and more vertebral fractures (GENOMOS, Ralston 2006).
LRP5	rs3736228 11:68433827 · C/T	CC	chip	T (0; 13%)	moderate	A1330V: T allele (1330V) reduces the Wnt signal in osteoblast; it is associated with lower mineral density and more fractures (GENOMOS, van Meurs 2008).
LRP5	rs4988321 11:68406721 · G/A/C	GG	chip	A (0; 4%)	moderate	V667M: Allele A (667M) is associated with less mineral density and more fractures (van Meurs 2008).
VDR	rs2228570 12:47879112 · A/C/G/T	GA	chip	A (1; 38%)	weak	FokI: the "f" allele (T in gene, A in strand +) gives a longer and slightly less active 3 amino acid receptor. Association with weak and inconsistent mineral density.
VDR	rs1544410 12:47846052 · C/A/G/T	CT	chip	T (1; 40%)	weak	Bsml: the "B" allele (A in gene, T in strand +). No association with mineral density or fractures in the GENOMOS consortium (Uitterlinden 2006).
VDR	rs731236 12:47844974 · A/G/T	AG	chip	G (1; 40%)	weak	TaqI: the allele "t" (C in gene, G in strand +), in linkage imbalance with Bsml "B". Weak evidence.
VDR	rs11568820 12:47908762 · C/T	CC	chip	C (2; 77%)	weak	Cdx2 (intestinal promotor): the G allele in the gene (C in strand +) gives less transcription of VDR in the intestine; the A allele was associated with ~9% less vertebral fractures (GENOMOS).
SOST	rs2741856 17:43749471 · G/C/T	GG	chip	G (2; 92%)	moderate	Sclerostine: inhibits the Wnt pathway in the bone; anti-sclerostine antibodies (romosozumab) are the strongest bone anabolic. Meaning: GWAS Catalog: Estimated mineral density (Chai RC 2026; p=3e-116); G ↓ allele
TNFRSF11B	rs7813157 8:118911126 · G/C/T	TT	chip	G (0; 5%)	moderate	Osteoprotegerin (OPG): a receptor-risk that blocks RANKL and stops bone resorption. Meaning: GWAS Catalog: Estimated mineral density (Chai RC 2026; p=2e-20); G ↓ allele
TNFSF11	rs117543324 13:42579733 · A/T	AA	imputation DR2=0.93	A (2; 97%)	moderate	RANKL: activates osteoclasts (target of denosumab). Meaning: GWAS Catalog: Estimated mineral density (Chai RC 2026; p=5e-28); allele A ↓
TNFRSF11A	rs884205 18:62387624 · A/C/T	CC	chip	A (0; 24%)	moderate	RANK: RANKL receptor in osteoclasts. Meaning: GWAS Catalog: Heel bone mineral density (Morris JA 2018; p=6e-38); allele A ↓
WNT16	rs2908007 7:121322110 · A/C/G	GA	chip	A (1; 60%)	moderate	WNT16: cortical thickness and cortical bone resistance. Meaning: GWAS Catalog: Estimated mineral density (Chai RC 2026; p=0e+00); allele A ↓

SPTBN1	rs4305309 2:54456574 · T/C	TC	imputation DR2=0.99	C (1; 64%)	moderate	Non-erythrocyte β -spectrin: mineral density locus and fractures. Meaning: GWAS Catalog: Estimated mineral density (Chai RC 2026; p=0e+00); allele T $\uparrow$
MEF2C	rs1366594 5:89080244 · A/C/G/T	AC	chip	C (1; 48%)	moderate	MEF2C: regulates sclerostine expression. Meaning: GWAS Catalog: Bone mineral density (hip) (Rivadeneira F 2009; p=1e-13); allele C $\downarrow$
ESR1	rs2982573 6:151689399 · T/C	CT	imputation DR2=1.00	T (1; 56%)	moderate	Estrogen receptor α : estrogen also maintains bone in men (via testosterone aromatization). Meaning: GWAS Catalog: Estimated mineral density (Chai RC 2026; p=0e+00); allele T $\downarrow$
JAG1	rs17457340 20:10659394 · T/C	TT	imputation DR2=0.99	C (0; 8%)	moderate	JAG1 (via Notch): mineral density locus and fractures. Meaning: GWAS Catalog: Heel bone mineral density (Kemp JP 2017; p=2e-25); allele T $\uparrow$
SOX6	rs56074687 11:16589928 · T/C	CT	imputation DR2=1.00	T (1; 72%)	moderate	SOX6: differentiation of chondrocytes and osteoblasts. Meaning: GWAS Catalog: Heel bone mineral density (Morris JA 2018; p=1e-33); allele T $\downarrow$

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FAM210A	rs4635400 — · —	—	SNP not found in the data	—	moderate	FAM210A: fracture locus and muscle strength (bone and muscle alloy). No major variant of GWAS Catalog for this gene with usable genotype and sense.

Articulations and tendons

Cartilage and tendons: GDF5 (the most replicated locus of osteoarthritis), type V collagen (COL5A1) and polygenic score of osteoarthritis. Tendons and cartilage lose collagen and elasticity with age and limit strength training in the long term.

Ageing mechanisms: Remodelling of the extracellular matrix; Stiffness of the extracellular matrix; Chronic Inflammation; Fragility

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Osteoarthritis	Favourable	the 16th percentile · $z=-0.97$	Higher score : higher risk. PGS001882 (PGS Catalog; Penalized regression (bigstatsr); Osteoarthritis; PGP000263, PMID 34995502). Variants with genotype: 11157 of 11451 (covering of 97.2% coefficients).
Tendon injuries (Aquiles, rotator cuff)	Typical / intermediate	the 72nd percentile · $z=0.52$	Higher score : higher risk. Few loci; low reliability. Loci score of GWAS Catalog ($p \leq 510^{-8}$; 1 variant by 500 kb window; $\ln(OR)$ or beta coefficients): 10 of 16 loci with genotype. Studies: Kim SK (Achilles tendinopathy); Kim SK (Achilles tendon injury); Kim SK (Shoulder impingement or rotator cuff loom); Roos TR (Rotator cuff injury); Tashjian RZ (Rotator cuff injury); Yanik EL (Degenerative rotator cuff disease) Variants with genotype: 10 of 16 (covering of 26.6% coefficients).

Favourable

- Osteoarthritis (the 16th percentile)
- GDF5 rs224333 AA

Unfavourable

- COL5A1 rs12722 TT

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
GDF5	rs224333 20:35436182 · A/G	AA	chip	G (0; 63%)	moderate	GDF5: cartilage and joint growth factor; the most replicated locus of osteoarthritis. Meaning: GWAS Catalog: Knee osteoarthritis (Hatzikotoulas K 2025; $p=6e-42$); allele A ↓

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
COL5A1	rs12722 9:134842570 · C/T	TT	chip	T (2; 58%)	weak	BstUI (3'UTR): T allele is associated with Achilles tendinopathy and ligament lesions in small studies; effect not confirmed in large GWAS.

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Fragility	Appendicular lean mass (the 10th percentile); Fractures (in the last 5 years) (the 84th percentile); ESR1 AA; RSPO3 AA; SMAD3 TT; COL5A1 TT	PPARGC1A CC; AMPD1 GG; ZBTB40 TC; AXIN1 CC; WLS CC; Osteoarthritis (the 16th percentile); GDF5 AA	Handgrip strength (mean of the two hands) (the 30th percentile); Normal gait speed (the 58th percentile); Sarcopenia (low grip strength at ≥ 60 years, EWGSOP/FNIH) (the 50th percentile); Frailty index (the 69th percentile); ACTN3 CT; Bone mineral density of the heel (ultrassom) (the 43rd percentile); Osteoporosis (the 42nd percentile); COL1A1 CC
Remodelling of the extracellular matrix	Fractures (in the last 5 years) (the 84th percentile); ESR1 AA; RSPO3 AA; SMAD3 TT; COL5A1 TT	ZBTB40 TC; AXIN1 CC; WLS CC; Osteoarthritis (the 16th percentile); GDF5 AA	Bone mineral density of the heel (ultrassom) (the 43rd percentile); Osteoporosis (the 42nd percentile); COL1A1 CC; LRP5 CC; LRP5 GG; SOST GG; TNFRSF11B TT; TNFSF11 AA
Decline of the Estrogen Signal	Fractures (in the last 5 years) (the 84th percentile); ESR1 AA; RSPO3 AA; SMAD3 TT	ZBTB40 TC; AXIN1 CC; WLS CC	Bone mineral density of the heel (ultrassom) (the 43rd percentile); Osteoporosis (the 42nd percentile); COL1A1 CC; LRP5 CC; LRP5 GG; SOST GG; TNFRSF11B TT; TNFSF11 AA
Endocrine dysregulation	Fractures (in the last 5 years) (the 84th percentile); ESR1 AA; RSPO3 AA; SMAD3 TT	ZBTB40 TC; AXIN1 CC; WLS CC	Bone mineral density of the heel (ultrassom) (the 43rd percentile); Osteoporosis (the 42nd percentile); COL1A1 CC; LRP5 CC; LRP5 GG; SOST GG; TNFRSF11B TT; TNFSF11 AA
Osteoporosis	Fractures (in the last 5 years) (the 84th percentile); ESR1 AA; RSPO3 AA; SMAD3 TT	ZBTB40 TC; AXIN1 CC; WLS CC	Bone mineral density of the heel (ultrassom) (the 43rd percentile); Osteoporosis (the 42nd percentile); COL1A1 CC; LRP5 CC; LRP5 GG; SOST GG; TNFRSF11B TT; TNFSF11 AA
Muscular atrophy	Appendicular lean mass (the 10th percentile)	PPARGC1A CC; AMPD1 GG	Handgrip strength (mean of the two hands) (the 30th percentile); Normal gait speed (the 58th percentile); Sarcopenia (low grip strength at ≥ 60 years, EWGSOP/FNIH) (the 50th percentile); Frailty index (the 69th percentile); ACTN3 CT
Oxidative Phosphorylation Decline	Appendicular lean mass (the 10th percentile)	PPARGC1A CC; AMPD1 GG	Handgrip strength (mean of the two hands) (the 30th percentile); Normal gait speed (the 58th percentile); Sarcopenia (low grip strength at ≥ 60 years, EWGSOP/FNIH) (the 50th percentile); Frailty index (the 69th percentile); ACTN3 CT
Neuromuscular Junction Disorder	Appendicular lean mass (the 10th percentile)	PPARGC1A CC; AMPD1 GG	Handgrip strength (mean of the two hands) (the 30th percentile); Normal gait speed (the 58th percentile); Sarcopenia (low grip strength at ≥ 60 years, EWGSOP/FNIH) (the 50th percentile); Frailty index (the 69th percentile); ACTN3 CT
Chronic Inflammation	COL5A1 TT	Osteoarthritis (percentile 16); GDF5 AA	Tendon injuries (Aquiles, rotator cuff) (the 72nd percentile)
Loss of muscle strength	Appendicular lean mass (the 10th percentile)	PPARGC1A CC; AMPD1 GG	Handgrip strength (mean of the two hands) (the 30th percentile); Normal gait speed (the 58th percentile); Sarcopenia (low grip strength at ≥ 60 years, EWGSOP/FNIH) (the 50th percentile); Frailty index (the 69th percentile); ACTN3 CT
Thin Mass Loss	Appendicular lean mass (the 10th percentile)	PPARGC1A CC; AMPD1 GG	Handgrip strength (mean of the two hands) (the 30th percentile); Normal gait speed (the 58th percentile); Sarcopenia (low grip strength at ≥ 60 years, EWGSOP/FNIH) (the 50th percentile); Frailty index (the 69th percentile); ACTN3 CT
Stiffness of the Extracellular Matrix	COL5A1 TT	Osteoarthritis (percentile 16); GDF5 AA	Tendon injuries (Aquiles, rotator cuff) (the 72nd percentile)
Sarcopenia	Appendicular lean mass (the 10th percentile)	PPARGC1A CC; AMPD1 GG	Handgrip strength (mean of the two hands) (the 30th percentile); Normal gait speed (the 58th percentile); Sarcopenia (low grip strength at ≥ 60 years, EWGSOP/FNIH) (the 50th percentile); Frailty index (the 69th percentile); ACTN3 CT

Practical part

Practical ideas taken only from the genetics of this report and crossed with the genetic plan v2 and current medication (isotretinoin, rapamycin 4 mg of 6 in 6 weeks, lithium 5 mg, tadalafil, dapagliflozin, retatrutide, telmisartan, LDN 4.5 mg, melatonin). The current stack remains as it is; nothing has been changed.

1. Progressive force training as a no. 1 priority

Genetics: Appendicular lean mass p9,5; grip strength p29,5; ACTN3 RX; p84 fractures

2–4 sessions per week of compound exercises (gaining, land survey, rowing, press), with load progression and 1–3 reserve repetitions. Add 1–2 short power blocks (fast rises, low jumps), because the power drops earlier than the force and ACTN3 RX benefits from specific training of fast fibers.

V2 genetic plan and medication: Coherent with the v2 plane (item 13: protecting lean mass under retatrutide). PPARGC1A Gly/Gly: also maintain aerobic training (2–3 zone sessions 2 + 1 intervals). • *suggestion (not applied)*

2. Impact load for the bone

Genetics: P84 fracture score; ESR1 XbaI AA, RSPO3 and SMAD3 unfavourable (weak)

The bone responds to high and fast loads: jumps (30–50 per session, 3×/week), short running or stairs, in addition to heavy force training. Balance and mobility 2×/week to prevent falls.

V2 genetic plan and medication: Compatible with the plan; without new supplements. • *suggestion (not applied)*

3. Distributed protein

Genetics: Thin mass p9, 5; typical sarcopenia (GRS p50)

Keep ~140 g/day in 4 meals of 30–40 g of complete protein, with a dose before sleeping. Leucine from food (fermented dairy, soybean, fish), not from BCAA isolates.

V2 genetic plan and medication: Similar to the v2 plane (protein ~140 g/day; BCAA isolates not recommended). • *suggestion (not applied)*

4. Creatine

Genetics: Thin mass p9,5; ACTN3 RX

Maintaining 5 g/day (also helps strength and, in the elderly, bone density when combined with training).

V2 genetic plan and medication: In the plane (S10, MANTER). • *suggestion (not applied)*

5. Vitamin D, K2 and calcium from food

Genetics: VDR FokI/BsmI/TaqI heterozygous (weak effects); p84 fractures

Maintaining D3 10 000 IU and K2 MK-7 240 µg; calcium comes mainly from food (fermented dairy products, tofu with calcium, green vegetables), with Ca-AKG at meals.

V2 genetic plan and medication: D3 10 000 IU and K2 as in the current protocol; Ca-AKG already in the plane (S14). • *suggestion (not applied)*

6. Collagen + vitamin C before training

Genetics: COL5A1 rs12722 TT (weak); p72 tendon lesion score

The 20 g of collagen can be taken 30–60 min before training; vitamin C is between meals (≥2 h away from food). Progressive warm-up and gradual increase in tendon exercises (Aquiles, rotuliano, shoulder).

V2 genetic plan and medication: Collagen 20 g (S9) and vitamin C between meals (S16) already in the plane. Note: the plan v2 describes the COL5A1 rs12722 TT as "no risk allele"; in the literature, the allele of risk for tendinopathy is T (weak evidence). • *suggestion (not applied)*

7. Omega-3 for protein synthesis

Genetics: Thin mass p9, 5

Omega-3 already taken (3.2–4 g/day) increases the anabolic response to protein in older adults.

V2 genetic plan and medication: Already taken; maintain. • *suggestion (not applied)*

8. Rapamycin and training

Genetics: mTORC1 is required for hypertrophy; lean mass p9,5

With 4 mg of 6 in 6 weeks, the effect on muscle is small. For caution, the heavier strength training can stay out of the day of taking and the following 2 days.

V2 genetic plan and medication: Compatible with the plan; weak evidence. • *suggestion (not applied)*

9. Retatrutide: protecting lean mass

Genetics: p9,5 lean mass; p29,5 force

Incretin agonists also reduce lean mass. Strength training + protein + creatine compensates most of the time. DEXA every 12 months to follow lean mass and bone density.

V2 genetic plan and medication: Similar to the 13 item of the v2 plane. • *suggestion (not applied)*

10. Medication with effect on bone or muscle

Genetics: p84 fractures

Lithium (even at a low dose): it is associated with fewer fractures (inhibes GSK-3 β and activates Wnt).

Telmisartan: blocking the AT1 receptor can attenuate muscle loss (weak evidence). Dapagliflozin: no sign of fractures (as opposed to canagliflozin). Isotretinoin: at high and long doses can cause muscle pain, CK rise with intense exercise and bone changes; with 40 mg/week the risk is low, but it is convenient to moisturize and not start very eccentric training blocks.

V2 genetic plan and medication: No changes to the medication. • *suggestion (not applied)*

Avoiding or with caution

- Preformed vitamin A (retinol) in supplements: added to isotretinoin and, in excess, increases the risk of fracture. *suggestion (not applied)*
- Excess of salt: increases the loss of calcium in urine (the profile already shows greater sensitivity to salt). *suggestion (not applied)*
- Proton pump inhibitors, corticosteroids and excess thyroid hormone without indication: worsen the bone. *suggestion (not applied)*
- Very intense eccentric training without progression (ACTN3 RX: more muscle damage), especially in phases of caloric deficit. *suggestion (not applied)*
- Fluorquinolones antibiotics, if there is an alternative (risk of tendon injury). *suggestion (not applied)*

Limitations

- Genotyping chip + local imputation (public reference panel). Variants imputed with $DR2 < 0.8$ were treated as not found. Indels (e.g. ECA I/D) are not read directly; a proxy is used.
- Polygenic scores explain a small fraction of the variation (about 10–20% for bone mineral density and less than 5% for force and fractures) and are not diagnostic. Training, protein, vitamin D and hormones weigh more.
- The percentiles compare with 633 people from the reference population. The scores were trained mainly in middle-aged adults and the elderly.
- The classical candidate genes (VDR, ESR1, COL5A1, MSTN) come from small studies; the effects are small and often not replicated. They are marked as weak evidence.
- The main variants by gene come from GWAS Catalog: the allele and the sense are those of the study with stronger association; some positions mapped to a gene can regulate neighboring genes.
- CAG repeats of the androgen receptor, structural variants and rare mutations (e.g., osteogenesis imperfecta) are not visible in the chip.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local imputation (Beagle 5.5 with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS001927 (PGS Catalog; Penalized regression (bigstatsr); Hand grip strength (mean of both); PGP000263, PMID 34995502 · PGS005315 (PGS Catalog; Genome-wide signifier SNPs; Appendicular lean mass; PGP000766, PMID 38632276 · PGS001075 (PGS Catalog; snpnet; Usual walking pace; PGP000244, PMID 35324888 · PGS001955 (PGS Catalog; Penalized regression (bigstatsr); Heel bone mineral density (BMD); PGP000263, PMID 34995502 · PGS001883 (PGS Catalog; Penalized regression (bigstatsr); Osteoporosis; PGP000263, PMID 34995502 · PGS001921 (PGS Catalog; Penalized regression (bigstatsr); Fractured/broken bones in last 5 years; PGP000263, PMID 34995502 · PGS001882 (PGS Catalog; Penalized regression (bigstatsr); Osteoarthritis; PGP000263, PMID 34995502).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentis: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

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